

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 10: Circles

CHAPTER 10: CIRCLES

A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

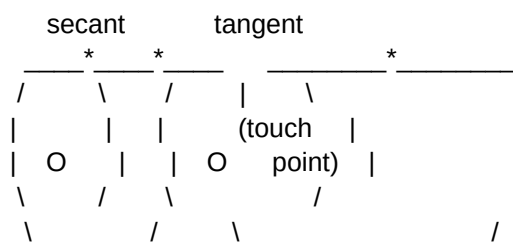
1.1 CIRCLE KI BASICS (RECAP)

- CIRCLE: ek point (centre) se same distance pe sabhi points ka set.
- RADIUS (r): centre se circle tak ki distance.
- DIAMETER (d): circle ke aar-paar (= 2 x radius).
- CHORD: circle pe do points ko jodne wali line.
- DIAMETER = sabse badi chord.

1.2 SECANT VS TANGENT (IS CHAPTER KA DIL)

SECANT: ek line jo circle ko DO points pe cut kare.

TANGENT: ek line jo circle ko SIRF EK point pe touch kare (cut nahi karti).



TANGENT ka touch point = "POINT OF CONTACT"

1.3 NUMBER OF TANGENTS

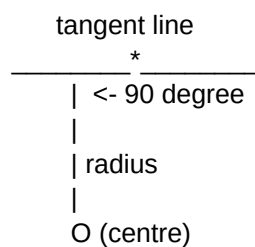
- Circle ke ANDAR ke point se: 0 tangents
- Circle PE wale point se: 1 tangent
- Circle ke BAAHAR point se: 2 tangents

SECTION 2: THEOREM 1 (TANGENT-RADIUS)

2.1 STATEMENT (BAHUT IMPORTANT)

"The tangent at any point of a circle is PERPENDICULAR to the radius through the point of contact."

MATLAB: Tangent aur radius (touch point pe) ke beech ka angle hamesha 90 degree hota hai.



2.2 USE

Jab bhi tangent involved ho, ek RIGHT-ANGLED triangle banta hai (radius + tangent + line to centre).
Phir Pythagoras laga sakte hain!

2.3 EXAMPLE

EXAMPLE 1: Ek circle ka radius 7 cm hai. Centre se 25 cm door ek point P hai. P se tangent ki length nikaal.

Tangent PT, radius OT = 7, OP = 25.
Angle OTP = 90 (tangent perp to radius).

Pythagoras in triangle OTP:

$$OP^2 = OT^2 + PT^2$$

$$25^2 = 7^2 + PT^2$$

$$625 = 49 + PT^2$$

$$PT^2 = 576$$

$$PT = 24 \text{ cm}$$

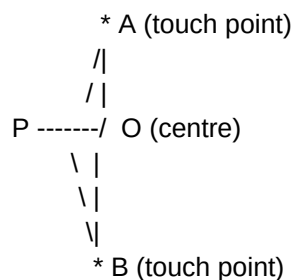
Tangent length = 24 cm (7-24-25 triplet!)

SECTION 3: THEOREM 2 (TWO TANGENTS)

3.1 STATEMENT (BAHUT IMPORTANT)

"The lengths of tangents drawn from an EXTERNAL point to a circle are EQUAL."

MATLAB: Circle ke bahar ke ek point se 2 tangents khinche, dono ki length BARABAR hogi.



$PA = PB$ (dono tangents equal)

3.2 EXTRA FACTS (from same theorem)

- $PA = PB$ (tangent lengths equal)
- Centre O point P ko join kare toh OP angle APB ko bisect karta hai (do barabar hisse)
- angle OPA = angle OPB

3.3 EXAMPLE

EXAMPLE 2: P se circle pe do tangents PA aur PB khinche. Agar $PA = 10$ cm, toh $PB = ?$

Theorem 2 se: $PB = PA = 10$ cm. (Equal tangents)

EXAMPLE 3: Do tangents 60 degree ka angle banate hain point P pe. Har tangent ka touch centre ke saath kya angle banayega?

angle APB = 60. OP bisect karta hai:
angle OPA = angle OPB = 30 degree.

SECTION 4: COMMON RESULTS

1. Tangent perpendicular to radius (90 degree).
2. Tangents from external point are equal.
3. Quadrilateral made by 2 radii + 2 tangents:
Opposite angles sum = 180 (cyclic-like result),
aur radius-tangent angles 90-90 hote hain.
4. Tangent + radius + centre line = right triangle
-> Pythagoras use karo.

Q AND A TIME

- Q1. Secant aur tangent mein kya difference hai?
- Q2. Circle ke bahar ke point se kitne tangents khinche ja sakte hain?
- Q3. Radius 8 cm. Centre se 17 cm door point P se tangent ki length nikaal. (Pythagoras)
- Q4. P se do tangents PA aur PB khinche. PA = 12 cm. PB kitna hoga? Kyun?
- Q5. Tangent aur radius ke beech (touch point pe) angle kitna hota hai?
- Q6. Do tangents 80 degree ka angle banate hain P pe. angle OPA (O = centre) kitna hoga?

SUMMARY

1. Secant: 2 points pe cut. Tangent: 1 point pe touch.
2. Tangents count:
inside point = 0, on circle = 1, outside = 2.
3. THEOREM 1: Tangent perpendicular to radius at point of contact (90 degree).

4. THEOREM 2: Tangents from external point are equal ($PA = PB$).
5. Tangent problems mein right triangle banta hai
-> Pythagoras laga.

TIPS FOR EXAM

1. Tangent + radius = 90 degree - ye yaad rakh, har tangent question mein use hota hai.
2. External point se equal tangents ($PA = PB$) -
bahut questions isi pe based hote hain.
3. Diagram banao aur 90 degree mark karo - phir Pythagoras ya geometry easily lag jaati hai.
4. Pythagorean triplets (7-24-25, 8-15-17, 3-4-5, 5-12-13) yaad rakh - calculation fast hoti hai.
5. Theorem statements word-to-word yaad rakh -
exam mein "state the theorem" aata hai.